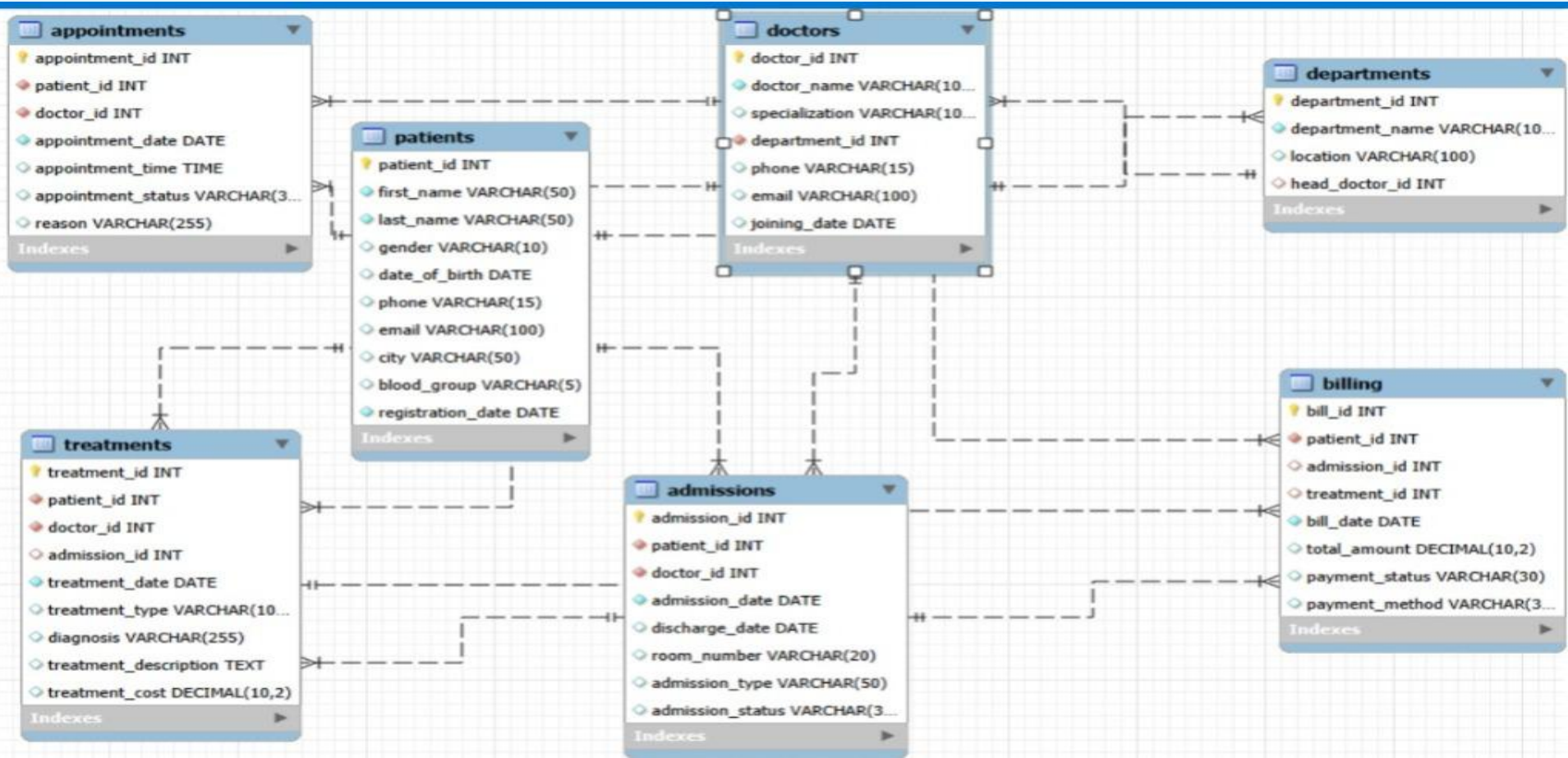
Hospital Mangement



Hospital Management Scope

The **Hospital Management Database System** is designed to manage hospital operations efficiently using a relational database. It stores and organizes information related to **departments, doctors, patients, appointments, admissions, treatments, and billing**. The database minimizes data redundancy, improves accuracy, and enables quick access to patient and hospital records. It uses **primary keys, foreign keys, joins, subqueries, views, and aggregate functions** to maintain data integrity and generate meaningful reports. The system supports appointment scheduling, patient treatment, billing management, and hospital analytics, providing a reliable and scalable solution for improving healthcare administration and decision-making processes.

ERR Diagram


appointments
appointment_id INT
patient_id INT
doctor_id INT
appointment_date DATE
appointment_time TIME
appointment_status VARCHAR(3...
reason VARCHAR(255)
Indexes
patients
patient_id INT
first_name VARCHAR(50)
last_name VARCHAR(50)
gender VARCHAR(10)
date_of_birth DATE
phone VARCHAR(15)
email VARCHAR(100)
city VARCHAR(50)
blood_group VARCHAR(5)
registration_date DATE
Indexes
doctors
doctor_id INT
doctor_name VARCHAR(10...
specialization VARCHAR(10...
department_id INT
phone VARCHAR(15)
email VARCHAR(100)
joining_date DATE
Indexes
departments
department_id INT
department_name VARCHAR(10...
location VARCHAR(100)
head_doctor_id INT
Indexes
treatments
treatment_id INT
patient_id INT
doctor_id INT
admission_id INT
treatment_date DATE
treatment_type VARCHAR(10...
diagnosis VARCHAR(255)
treatment_description TEXT
treatment_cost DECIMAL(10,2)
Indexes
admissions
admission_id INT
patient_id INT
doctor_id INT
admission_date DATE
discharge_date DATE
room_number VARCHAR(20)
admission_type VARCHAR(50)
admission_status VARCHAR(3...
Indexes
billing
bill_id INT
patient_id INT
admission_id INT
treatment_id INT
bill_date DATE
total_amount DECIMAL(10,2)
payment_status VARCHAR(30)
payment_method VARCHAR(3...
Indexes

TABLES STRUTURE

Structure of Department Table

```
6 •  desc departments;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
department_id	int	NO	PRI	NULL	auto_increment
department_name	varchar(100)	NO	UNI	NULL	
location	varchar(100)	YES		NULL	
head_doctor_id	int	YES	MUL	NULL	

Department table stores hospital department details, including department ID, name, location and assigned head doctor.

Syntax: desc departments;

Structure of Patient Table

```
8 •  DESC patients;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
patient_id	int	NO	PRI	NULL	auto_increment
first_name	varchar(50)	NO		NULL	
last_name	varchar(50)	NO		NULL	
gender	varchar(10)	YES		NULL	
date_of_birth	date	YES		NULL	
phone	varchar(15)	YES		NULL	
email	varchar(100)	YES		NULL	
city	varchar(50)	YES		NULL	
blood_group	varchar(5)	YES		NULL	
registration_date	date	NO		NULL	

The **Patient** table stores patient information, including patient ID, personal details, contact information, gender, age, city, registration date and etc.

Syntax: DESC patients;

Structure of Doctor Table

```
10 •  DESC doctors;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conter

Field	Type	Null	Key	Default	Extra
doctor_id	int	NO	PRI	NULL	auto_increment
doctor_name	varchar(100)	NO		NULL	
specialization	varchar(100)	YES		NULL	
department_id	int	NO	MUL	NULL	
phone	varchar(15)	YES		NULL	
email	varchar(100)	YES		NULL	
joining_date	date	YES		NULL	

The **Doctors** table stores doctor information including doctor ID, name, specialization, department, contact details and joining date.

Syntax: DESC doctors;

Structure of Appointment Table

```
12 •  DESC Appointments;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
appointment_id	int	NO	PRI	NULL	auto_increment
patient_id	int	NO	MUL	NULL	
doctor_id	int	NO	MUL	NULL	
appointment_date	date	NO		NULL	
appointment_time	time	YES		NULL	
appointment_status	varchar(30)	YES		NULL	
reason	varchar(255)	YES		NULL	

The **Appointments** table stores appointment details including appointment ID, patient ID, doctor ID, appointment date, time, and status.

Syntax: DESC Appointments;

Structure of Admission Table

```
12 •  DESC admissions;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conter

Field	Type	Null	Key	Default	Extra
admission_id	int	NO	PRI	NULL	auto_increment
patient_id	int	NO	MUL	NULL	
doctor_id	int	NO	MUL	NULL	
admission_date	date	NO		NULL	
discharge_date	date	YES		NULL	
room_number	varchar(20)	YES		NULL	
admission_type	varchar(50)	YES		NULL	
admission_status	varchar(30)	YES		NULL	

The **Admissions** table stores patient admission details including admission ID, patient ID, doctor ID, admission date, discharge date, admission type ,status and room number..

Syntax: DESC admissions;

Structure of Treatment Table

```
14 •  DESC treatments;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
treatment_id	int	NO	PRI	NULL	auto_increment
patient_id	int	NO	MUL	NULL	
doctor_id	int	NO	MUL	NULL	
admission_id	int	YES	MUL	NULL	
treatment_date	date	NO		NULL	
treatment_type	varchar(100)	YES		NULL	
diagnosis	varchar(255)	YES		NULL	
treatment_description	text	YES		NULL	
treatment_cost	decimal(10,2)	YES		NULL	

The **Treatments** table stores treatment details including treatment ID, patient ID, doctor ID, admission ID, treatment description, diagnosis, treatment type, treatment cost, and date.

Syntax: DESC treatments;

Structure of Billing Table

```
16 •   DESC BILLING;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
bill_id	int	NO	PRI	NULL	auto_increment
patient_id	int	NO	MUL	NULL	
admission_id	int	YES	MUL	NULL	
treatment_id	int	YES	MUL	NULL	
bill_date	date	NO		NULL	
total_amount	decimal(10,2)	YES		NULL	
payment_status	varchar(30)	YES		NULL	
payment_method	varchar(30)	YES		NULL	

The **Billing** table stores billing information including bill ID, patient ID, treatment ID, total amount, payment method, and payment status.

Syntax: DESC BILLING;

TABLES CONTENT

Content of Department Table

```
18 •    SELECT * FROM Departments;
```

Result Grid | Filter Rows: | Edit:

department_id	department_name	location	head_doctor_id
1	Cardiology	Block A - 1st Floor	NULL
2	Neurology	Block A - 2nd Floor	NULL
3	Orthopedics	Block B - 1st Floor	NULL
4	Pediatrics	Block B - 2nd Floor	NULL
5	Dermatology	Block C - 1st Floor	NULL
6	General Medicine	Block C - 2nd Floor	NULL
7	Gynecology	Block D - 1st Floor	NULL
8	Oncology	Block D - 2nd Floor	NULL
9	Emergency Medicine	Emergency Block	NULL
10	ENT	Block E - 1st Floor	NULL
11	Psychiatry	Block E - 2nd Floor	NULL
12	Urology	Block F - 1st Floor	NULL
13	Nephrology	Block F - 2nd Floor	NULL
14	Gastroenterology	Block G - 1st Floor	NULL
15	Pulmonology	Block G - 2nd Floor	NULL
16	Endocrinology	Block H - 1st Floor	NULL

Content of Patient Table

```
20 •    SELECT * FROM Patients;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

patient_id	first_name	last_name	gender	date_of_birth	phone	email	city	blood_group	registration_d
1	Aarav	Sharma	Male	1985-03-15	9876501001	aarav.sharma@gmail.com	Mumbai	O+	2023-01-10
2	Priya	Patil	Female	1992-07-22	9876501002	priya.patil@gmail.com	Pune	A+	2023-01-15
3	Rohan	Mehta	Male	1978-11-05	9876501003	rohan.mehta@gmail.com	Mumbai	B+	2023-02-03
4	Sneha	Joshi	Female	1995-01-18	9876501004	sneha.joshi@gmail.com	Nashik	AB+	2023-02-12
5	Vikram	Deshmukh	Male	1969-06-30	9876501005	vikram.deshmukh@gmail.com	Nagpur	O-	2023-02-20
6	Anjali	Kulkarni	Female	1988-09-12	9876501006	anjali.kulkarni@gmail.com	Mumbai	B-	2023-03-01
7	Rahul	Shah	Male	1990-04-25	9876501007	rahul.shah@gmail.com	Thane	A-	2023-03-10
8	Neha	Verma	Female	1982-12-08	9876501008	neha.verma@gmail.com	Pune	O+	2023-03-18
9	Amit	Rao	Male	1975-08-19	9876501009	amit.rao@gmail.com	Mumbai	AB-	2023-03-25
10	Pooja	Nair	Female	1998-05-14	9876501010	pooja.nair@gmail.com	Navi Mumbai	A+	2023-04-02
11	Sanjay	Gupta	Male	1965-10-27	9876501011	sanjay.gupta@gmail.com	Mumbai	B+	2023-04-15
12	Kavita	Iyer	Female	1987-02-11	9876501012	kavita.iyer@gmail.com	Pune	O+	2023-04-22
13	Akash	Tiwari	Male	1993-06-17	9876501013	akash.tiwari@gmail.com	Aurangabad	A-	2023-05-05
14	Riya	Bansal	Female	1991-11-23	9876501014	riya.bansal@gmail.com	Mumbai	AB+	2023-05-12
15	Manish	Yadav	Male	1972-03-09	9876501015	manish.yadav@gmail.com	Thane	O-	2023-05-20

Content of Doctor Table

```
22 •    SELECT * FROM doctors;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

doctor_id	doctor_name	specialization	department_id	phone	email	joining_date
1	Dr. Rajesh Sharma	Cardiologist	1	9876543201	rajesh.sharma@hospital.com	2018-04-12
2	Dr. Priya Mehta	Neurologist	2	9876543202	priya.mehta@hospital.com	2019-07-15
3	Dr. Amit Verma	Orthopedic Surgeon	3	9876543203	amit.verma@hospital.com	2017-02-20
4	Dr. Neha Kapoor	Pediatrician	4	9876543204	neha.kapoor@hospital.com	2020-01-10
5	Dr. Rohit Joshi	Dermatologist	5	9876543205	rohit.joshi@hospital.com	2018-09-18
6	Dr. Sneha Patil	General Physician	6	9876543206	sneha.patil@hospital.com	2019-03-25
7	Dr. Anjali Deshmukh	Gynecologist	7	9876543207	anjali.deshmukh@hospital.com	2016-11-05
8	Dr. Vikram Malhotra	Oncologist	8	9876543208	vikram.malhotra@hospital.com	2015-06-14
9	Dr. Karan Singh	Emergency Physician	9	9876543209	karan.singh@hospital.com	2021-08-22
10	Dr. Meera Nair	ENT Specialist	10	9876543210	meera.nair@hospital.com	2017-12-01
11	Dr. Arjun Rao	Psychiatrist	11	9876543211	arjun.rao@hospital.com	2019-05-17
12	Dr. Pooja Kulkarni	Urologist	12	9876543212	pooja.kulkarni@hospital.com	2018-10-11
13	Dr. Suresh Iyer	Nephrologist	13	9876543213	suresh.iyer@hospital.com	2016-03-08
14	Dr. Kavita Shah	Gastroenterologist	14	9876543214	kavita.shah@hospital.com	2020-06-19
15	Dr. Nitin Gupta	Pulmonologist	15	9876543215	nitin.gupta@hospital.com	2017-08-27

Content of Appointment Table

```
24 •    SELECT * FROM appointments;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

appointment_id	patient_id	doctor_id	appointment_date	appointment_time	appointment_status	reason
1	1	1	2023-01-12	09:30:00	Completed	Chest pain and routine cardiac checkup
2	2	7	2023-01-18	10:00:00	Completed	Routine gynecological consultation
3	3	2	2023-02-05	11:30:00	Completed	Frequent headaches and dizziness
4	4	4	2023-02-14	09:00:00	Completed	Child vaccination and health checkup
5	5	3	2023-02-22	14:00:00	Completed	Knee pain and joint stiffness
6	6	5	2023-03-03	10:30:00	Completed	Skin allergy and rash
7	7	6	2023-03-12	11:00:00	Completed	Fever and general weakness
8	8	15	2023-03-20	15:30:00	Completed	Breathing difficulty and cough
9	9	8	2023-03-27	09:30:00	Completed	Oncology consultation
10	10	10	2023-04-04	10:00:00	Completed	Ear pain and hearing problem
11	11	11	2023-04-17	13:00:00	Completed	Anxiety and sleep problems
12	12	13	2023-04-24	11:30:00	Completed	Kidney function consultation
13	13	14	2023-05-07	09:00:00	Completed	Stomach pain and digestive issues
14	14	16	2023-05-14	14:30:00	Completed	Diabetes and hormone checkup
15	15	17	2023-05-22	10:30:00	Completed	Joint pain and inflammation

Content of Admission Table

```
26 •    SELECT * FROM admissions;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

admission_id	patient_id	doctor_id	admission_date	discharge_date	room_number	admission_type	admission_status
1	1	1	2023-01-13	2023-01-18	A101	Emergency	Discharged
2	2	7	2023-01-20	2023-01-23	D201	Scheduled	Discharged
3	3	2	2023-02-06	2023-02-12	A205	Emergency	Discharged
4	4	4	2023-02-15	2023-02-17	B102	Scheduled	Discharged
5	5	3	2023-02-23	2023-03-02	B105	Emergency	Discharged
6	6	5	2023-03-04	2023-03-06	C101	Scheduled	Discharged
7	7	6	2023-03-13	2023-03-16	C205	Emergency	Discharged
8	8	15	2023-03-21	2023-03-28	G202	Emergency	Discharged
9	9	8	2023-03-28	2023-04-05	D205	Scheduled	Discharged
10	10	10	2023-04-05	2023-04-07	E101	Scheduled	Discharged
11	11	11	2023-04-18	2023-04-25	E205	Emergency	Discharged
12	12	13	2023-04-25	2023-05-02	F201	Scheduled	Discharged
13	13	14	2023-05-08	2023-05-13	G101	Emergency	Discharged
14	14	16	2023-05-15	2023-05-20	H101	Scheduled	Discharged
15	15	17	2023-05-23	2023-05-29	H205	Emergency	Discharged

Content of Treatment Table

```
28 •  SELECT * FROM treatments;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

treatment_id	patient_id	doctor_id	admission_id	treatment_date	treatment_type	diagnosis	treatment_description	treatment_cost
1	1	1	1	2023-01-14	Cardiac Evaluation	Hypertension	ECG, blood pressure monitoring and cardiac ass...	8500.00
2	2	7	2	2023-01-21	Gynecological Treatment	Ovarian Cyst	Ultrasound examination and medication-based t...	12000.00
3	3	2	3	2023-02-07	Neurological Treatment	Migraine	Neurological examination and migraine manage...	9500.00
4	4	4	4	2023-02-16	Pediatric Treatment	Viral Fever	Medical observation, medication and hydration t...	5000.00
5	5	3	5	2023-02-24	Orthopedic Treatment	Knee Arthritis	X-ray, physiotherapy and pain management	18000.00
6	6	5	6	2023-03-05	Dermatology Treatment	Skin Allergy	Allergy testing and prescribed skin treatment	6500.00
7	7	6	7	2023-03-14	General Medicine	Dengue Fever	Blood tests, hydration therapy and medical mon...	15000.00
8	8	15	8	2023-03-22	Pulmonary Treatment	Asthma	Pulmonary function test and respiratory treatment	11000.00
9	9	8	9	2023-03-29	Cancer Treatment	Breast Cancer	Cancer evaluation and chemotherapy planning	75000.00
10	10	10	10	2023-04-06	ENT Treatment	Ear Infection	Ear examination, cleaning and antibiotic treatment	5500.00
11	11	11	11	2023-04-19	Psychiatric Treatment	Anxiety Disor...	Psychiatric evaluation and counseling sessions	9000.00
12	12	13	12	2023-04-26	Kidney Treatment	Chronic Kidne...	Kidney function monitoring and medication man...	25000.00
13	13	14	13	2023-05-09	Gastrointestinal Treatm...	Gastritis	Endoscopy and medication-based treatment	14000.00
14	14	16	14	2023-05-16	Endocrine Treatment	Type 2 Diabe...	Blood sugar monitoring and diabetes management	10000.00
15	15	17	15	2023-05-24	Rheumatology Treatment	Rheumatoid ...	Joint evaluation and anti-inflammatory treatment	16000.00

Content of Billing Table

```
30 •  SELECT * FROM Billing;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

bill_id	patient_id	admission_id	treatment_id	bill_date	total_amount	payment_status	payment_method
1	1	1	1	2023-01-18	10500.00	Paid	Cash
2	2	2	2	2023-01-23	14500.00	Paid	Credit Card
3	3	3	3	2023-02-12	11000.00	Paid	UPI
4	4	4	4	2023-02-17	6000.00	Paid	Cash
5	5	5	5	2023-03-02	22000.00	Paid	Debit Card
6	6	6	6	2023-03-06	7500.00	Paid	UPI
7	7	7	7	2023-03-16	18000.00	Paid	Insurance
8	8	8	8	2023-03-28	13500.00	Paid	Credit Card
9	9	9	9	2023-04-05	85000.00	Pending	Insurance
10	10	10	10	2023-04-07	6500.00	Paid	UPI
11	11	11	11	2023-04-25	10500.00	Paid	Debit Card
12	12	12	12	2023-05-02	28000.00	Pending	Insurance
13	13	13	13	2023-05-13	16500.00	Paid	Cash
14	14	14	14	2023-05-20	12000.00	Paid	UPI
15	15	15	15	2023-05-29	19000.00	Paid	Credit Card

SUBQUIRES

Single-Row Subquery

❖ Find the treatment with the highest cost.

SYNTAX: SELECT Patient_id, treatment_type, treatment_date, treatment_cost FROM Treatments WHERE treatment_cost =(SELECT MAX(treatment_cost) FROM Treatments);

```
38      --  Find the treatment with the highest cost.(Single-Row Subquerie)
39  •   SELECT Patient_id,treatment_type,treatment_date,treatment_cost FROM Treatments WHERE treatment_cost =
40      (SELECT MAX(treatment_cost) FROM Treatments);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Patient_id	treatment_type	treatment_date	treatment_cost
25	Critical Care	2023-08-24	120000.00

Single-Row Subquery

❖ Find doctors working in the Cardiology department.

SYNTAX: SELECT * FROM Doctors WHERE department_id =(SELECT department_id FROM Departments WHERE department_name='Cardiology');

```
39    -- Find doctors working in the Cardiology department. (Single-Row Subquerie)
40 •  SELECT * FROM Doctors WHERE department_id =
41    (SELECT department_id FROM Departments WHERE department_name='Cardiology');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

doctor_id	doctor_name	specialization	department_id	phone	email	joining_date
1	Dr. Rajesh Sharma	Cardiologist	1	9876543201	rajesh.sharma@hospital.com	2018-04-12
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Multi-Row Subquery

❖ Find doctors who belong to departments located in Block A.

SYNTAX: SELECT doctor_id, doctor_name, specialization, department_id FROM Doctors WHERE department_id IN(SELECT department_id FROM Departments WHERE location LIKE 'Block A%');

```
43      -- Find doctors who belong to departments located in Block A. (Multiple-Row Subquerie)
44  •   SELECT doctor_id,doctor_name,specialization,department_id FROM Doctors WHERE department_id IN
45      ( SELECT department_id FROM Departments WHERE location LIKE 'Block A%');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

doctor_id	doctor_name	specialization	department_id
1	Dr. Rajesh Sharma	Cardiologist	1
2	Dr. Priya Mehta	Neurologist	2
NULL	NULL	NULL	NULL

Multi-Row Subquery

- Find patients having pending bills.

SYNTAX: SELECT * FROM Patients WHERE patient_id IN(SELECT patient_id FROM Billing WHERE payment_status='Pending');

```
47      -- Find patients having pending bills. (Multiple-Row Subquerie)
48  •   SELECT * FROM Patients WHERE patient_id IN
49      (SELECT patient_id FROM Billing WHERE payment_status='Pending' );
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

patient_id	first_name	last_name	gender	date_of_birth	phone	email	city	blood_group	registration_date
9	Amit	Rao	Male	1975-08-19	9876501009	amit.rao@gmail.com	Mumbai	AB-	2023-03-25
12	Kavita	Iyer	Female	1987-02-11	9876501012	kavita.iyer@gmail.com	Pune	O+	2023-04-22
16	Swati	Desai	Female	1985-07-28	9876501016	swati.desai@gmail.com	Nashik	B+	2023-06-01
20	Isha	Agarwal	Female	1994-12-13	9876501020	isha.agarwal@gmail.com	Mumbai	AB+	2023-07-03
21	Deepak	Mishra	Male	1980-08-07	9876501021	deepak.mishra@gmail.com	Nagpur	O+	2023-07-15
25	Mohan	Reddy	Male	1963-12-02	9876501025	mohan.reddy@gmail.com	Mumbai	AB-	2023-08-20
29	Aditya	Joshi	Male	1999-11-30	9876501029	aditya.joshi@gmail.com	Pune	A+	2023-10-05
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Correlated Subquery

- Find doctors whose average treatment cost is greater than ₹50,000

SYNTAX: select d. doctor_name FROM Doctors d WHERE (SELECT AVG(t.treatment_cost) FROM Treatments t WHERE t.doctor_id = d.doctor_id) > 50000;

```
51    -- Find doctors whose average treatment cost is greater than ₹50,000 (Correlated Subquerie)
52    select d. doctor_name FROM Doctors d
53    WHERE (SELECT AVG(t.treatment_cost) FROM Treatments t WHERE t.doctor_id = d.doctor_id) > 50000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

doctor_name
Dr. Vikram Malhotra
Dr. Isha Agarwal
Dr. Sameer Khan
Dr. Sanjay Kulkarni

JOINS

Inner Join

- Display billing with doctor names.

SYNTAX: SELECT b.patient_id, d.doctor_name, b.total_amount, b.payment_status FROM Billing b
INNER JOIN Treatments t ON b.treatment_id = t.treatment_id
INNER JOIN Doctors d ON t.doctor_id = d.doctor_id;

```
60      -- Display billing with doctor names (Inner Join)
61  •   SELECT b.patient_id, d.doctor_name, b.total_amount, b.payment_status FROM Billing b
62      INNER JOIN Treatments t ON b.treatment_id = t.treatment_id
63      INNER JOIN Doctors d ON t.doctor_id = d.doctor_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

patient_id	doctor_name	total_amount	payment_status
1	Dr. Rajesh Sharma	10500.00	Paid
2	Dr. Anjali Deshmukh	14500.00	Paid
3	Dr. Priya Mehta	11000.00	Paid
4	Dr. Neha Kapoor	6000.00	Paid
5	Dr. Amit Verma	22000.00	Paid
6	Dr. Rohit Joshi	7500.00	Paid
7	Dr. Sneha Patil	18000.00	Paid
8	Dr. Nitin Gupta	13500.00	Paid
9	Dr. Vikram Malhotra	85000.00	Pending
10	Dr. Meera Nair	6500.00	Paid
11	Dr. Arjun Rao	10500.00	Paid
12	Dr. Suresh Iyer	28000.00	Pending
13	Dr. Kavita Shah	16500.00	Paid

Left Join

- Display all departments with doctors and appointments

SYNTAX: SELECT dep.department_name, d.doctor_name, a.appointment_date, a.appointment_time, d.specialization FROM Departments dep
LEFT JOIN Doctors d ON dep.department_id = d.department_id
LEFT JOIN Appointments a ON d.doctor_id = a.doctor_id;

```
65      -- Display all departments with doctors and appointments(Left Join)
66  •   SELECT dep.department_name, d.doctor_name, a.appointment_date, a.appointment_time,d.specialization
67      FROM Departments dep
68      LEFT JOIN Doctors d ON dep.department_id = d.department_id
69      LEFT JOIN Appointments a ON d.doctor_id = a.doctor_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_name	doctor_name	appointment_date	appointment_time	specialization
Anesthesiology	Dr. Nisha Reddy	2023-06-27	11:00:00	Anesthesiologist
Cardiology	Dr. Rajesh Sharma	2023-01-12	09:30:00	Cardiologist
Critical Care	Dr. Sanjay Kulkarni	2023-08-22	16:00:00	Critical Care Specialist
Dentistry	Dr. Deepak Mishra	2023-06-12	15:00:00	Dentist
Dermatology	Dr. Rohit Joshi	2023-03-03	10:30:00	Dermatologist
Emergency Medicine	Dr. Karan Singh	2023-09-20	18:00:00	Emergency Physician
Endocrinology	Dr. Riya Bansal	2023-05-14	14:30:00	Endocrinologist
ENT	Dr. Meera Nair	2023-04-04	10:00:00	ENT Specialist
Gastroenterology	Dr. Kavita Shah	2023-05-07	09:00:00	Gastroenterologist
General Medicine	Dr. Sneha Patil	2023-03-12	11:00:00	General Physician
General Surgery	Dr. Rahul Chavan	2023-07-05	14:00:00	General Surgeon
Gynecology	Dr. Anjali Deshmukh	2023-01-18	10:00:00	Gynecologist
Hematology	Dr. Aarti Choudhary	2023-08-07	09:00:00	Hematologist
Infectious Diseases	Dr. Mohan Reddy	2023-07-24	13:30:00	Infectious Disease S...
Nephrology	Dr. Suresh Iyer	2023-04-24	11:30:00	Nephrologist

Right Join

- Display all appointments with department names

SYNTAX: SELECT dep.department_name, d.doctor_name, a.appointment_date FROM Departments dep
RIGHT JOIN Doctors d ON dep.department_id = d.department_id
RIGHT JOIN Appointments a ON d.doctor_id = a.doctor_id;

```
71      -- Display all appointments with department names (Right Join)
72      SELECT dep.department_name, d.doctor_name, a.appointment_date FROM Departments dep
73      RIGHT JOIN Doctors d ON dep.department_id = d.department_id
74      RIGHT JOIN Appointments a ON d.doctor_id = a.doctor_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_name	doctor_name	appointment_date
Cardiology	Dr. Rajesh Sharma	2023-01-12
Gynecology	Dr. Anjali Deshmukh	2023-01-18
Neurology	Dr. Priya Mehta	2023-02-05
Pediatrics	Dr. Neha Kapoor	2023-02-14
Orthopedics	Dr. Amit Verma	2023-02-22
Dermatology	Dr. Rohit Joshi	2023-03-03
General Medicine	Dr. Sneha Patil	2023-03-12
Pulmonology	Dr. Nitin Gupta	2023-03-20
Oncology	Dr. Vikram Malhotra	2023-03-27
ENT	Dr. Meera Nair	2023-04-04
Psychiatry	Dr. Arjun Rao	2023-04-17
Nephrology	Dr. Suresh Iyer	2023-04-24
Gastroenterology	Dr. Kavita Shah	2023-05-07
Endocrinology	Dr. Riya Bansal	2023-05-14
Rheumatology	Dr. Manish Yadav	2023-05-22

Union Join

- List doctor names and patient names alphabetically

SYNTAX SELECT doctor_name AS doc_Name FROM Doctors
UNION
SELECT CONCAT(first_name,' ',last_name) FROM Patients ORDER BY doc_Name;

```
76      -- List doctor names and patient names alphabetically (Union)
77 •    SELECT doctor_name AS doc_Name FROM Doctors UNION
78      SELECT CONCAT(first_name,' ',last_name) FROM Patients ORDER BY doc_Name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

doc_Name
Aarav Sharma
Aarti Choudhary
Aditya Joshi
Akash Tiwari
Amit Rao
Anjali Kulkarni
Deepak Mishra
Dr. Aarti Choudhary
Dr. Akash Tiwari
Dr. Amit Verma
Dr. Anjali Deshmukh
Dr. Arjun Rao
Dr. Deepak Mishra
Dr. Isha Agarwal

Self Join

❖ Patients with the same blood group

SYNTAX : SELECT p1.first_name AS Patient1, p2.first_name AS Patient2, p1.blood_group FROM Patients p1, Patients p2
WHERE p1.blood_group = p2.blood_group AND p1.patient_id < p2.patient_id;

```
91      --  Patients with the same blood group (Self Join)
92  •   SELECT p1.first_name AS Patient1, p2.first_name AS Patient2, p1.blood_group FROM Patients p1, Patients p2
93      WHERE p1.blood_group = p2.blood_group AND p1.patient_id < p2.patient_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Patient1	Patient2	blood_group
Aarav	Neha	O+
Aarav	Kavita	O+
Aarav	Meera	O+
Aarav	Deepak	O+
Aarav	Monica	O+
Priya	Pooja	A+
Priya	Karan	A+
Priya	Nisha	A+
Priya	Aditya	A+
Rohan	Sanjay	B+
Rohan	Swati	B+
Rohan	Sameer	B+
Rohan	Aarti	B+
Sneha	Riya	AB+

VIEWS

VIEWS

❖ Patient Billing View

SYNTAX : CREATE VIEW Patient_Billing_View AS
SELECT p.patient_id, CONCAT(p.first_name,' ',p.last_name) AS patient_name, b.total_amount, b.payment_status, b.payment_method FROM Patients p
JOIN Billing b ON p.patient_id = b.patient_id;

patient_id	patient_name	total_amount	payment_status	payment_method
1	Aarav Sharma	10500.00	Paid	Cash
2	Priya Patil	14500.00	Paid	Credit Card
3	Rohan Mehta	11000.00	Paid	UPI
4	Sneha Joshi	6000.00	Paid	Cash
5	Vikram Deshmukh	22000.00	Paid	Debit Card
6	Anjali Kulkarni	7500.00	Paid	UPI
7	Rahul Shah	18000.00	Paid	Insurance
8	Neha Verma	13500.00	Paid	Credit Card
9	Amit Rao	85000.00	Pending	Insurance
10	Pooja Nair	6500.00	Paid	UPI
11	Sanjay Gupta	10500.00	Paid	Debit Card
12	Kavita Iyer	28000.00	Pending	Insurance
13	Akash Tiwari	16500.00	Paid	Cash
14	Riya Bansal	12000.00	Paid	UPI
15	Manish Yadav	19000.00	Paid	Credit Card

VIEWS

❖ Emergency Admissions View

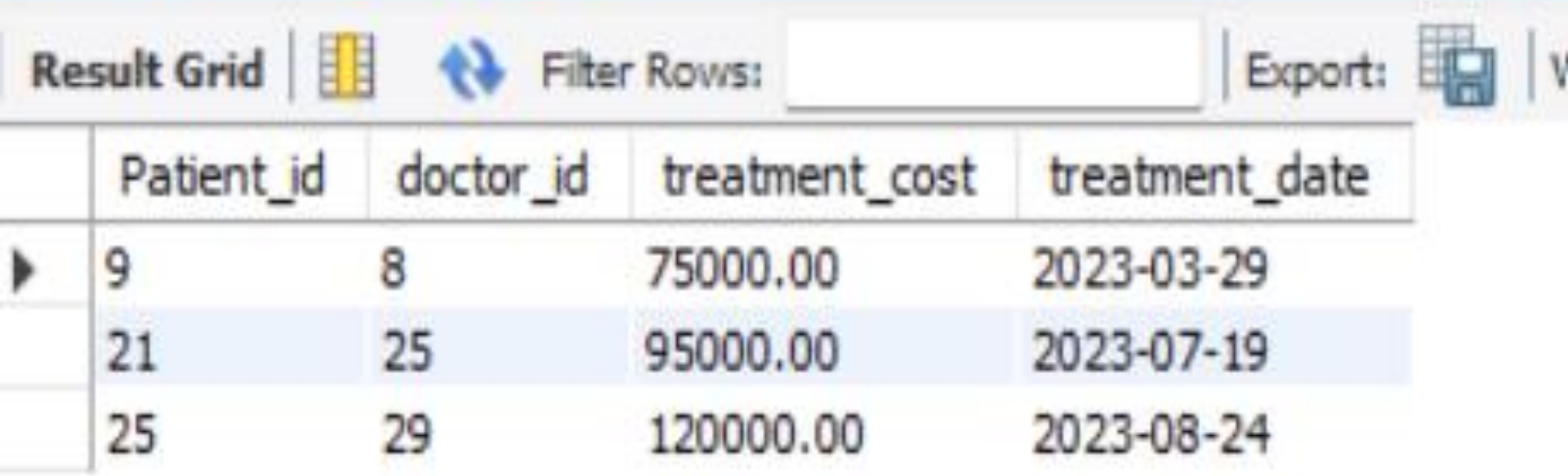
SYNTAX : CREATE VIEW Emergency_Admissions AS
SELECT patient_id, doctor_id, admission_type, admission_status FROM Admissions WHERE admission_type='Emergency';

patient_id	doctor_id	admission_type	admission_status
1	1	Emergency	Discharged
3	2	Emergency	Discharged
5	3	Emergency	Discharged
7	6	Emergency	Discharged
8	15	Emergency	Discharged
11	11	Emergency	Discharged
13	14	Emergency	Discharged
15	17	Emergency	Discharged
19	22	Emergency	Discharged
21	25	Emergency	Discharged
22	27	Emergency	Discharged
25	29	Emergency	Admitted
28	9	Emergency	Discharged

VIEWS

❖ High Cost Treatments View

SYNTAX : CREATE VIEW High_Cost_Treatments AS
SELECT Patient_id, doctor_id, treatment_cost, treatment_date FROM Treatments WHERE treatment_cost > 60000;



Patient_id	doctor_id	treatment_cost	treatment_date
9	8	75000.00	2023-03-29
21	25	95000.00	2023-07-19
25	29	120000.00	2023-08-24

VIEWS

❖ Paid Bills View

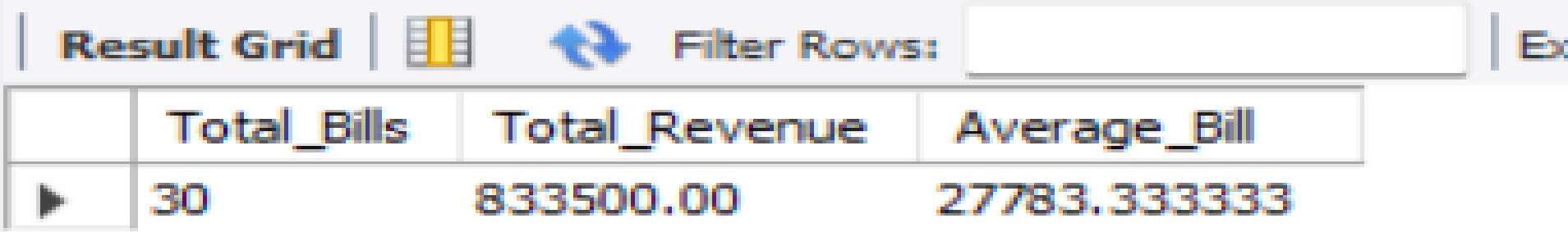
SYNTAX : CREATE VIEW Paid_Bills AS
SELECT patient_id, total_amount, payment_status, payment_method FROM Billing WHERE payment_status='Paid';

patient_id	total_amount	payment_status	payment_method
1	10500.00	Paid	Cash
2	14500.00	Paid	Credit Card
3	11000.00	Paid	UPI
4	6000.00	Paid	Cash
5	22000.00	Paid	Debit Card
6	7500.00	Paid	UPI
7	18000.00	Paid	Insurance
8	13500.00	Paid	Credit Card
10	6500.00	Paid	UPI
11	10500.00	Paid	Debit Card
13	16500.00	Paid	Cash
14	12000.00	Paid	UPI
15	19000.00	Paid	Credit Card
17	6000.00	Paid	Cash
18	14500.00	Paid	Debit Card

VIEWS

❖ Total Hospital Billing Summary

SYNTAX : CREATE VIEW Hospital_Billing_Summary AS
SELECT COUNT(*) AS Total_Bills, SUM(total_amount) AS Total_Revenue, AVG(total_amount) AS Average_Bill
FROM Billing;



Total_Bills	Total_Revenue	Average_Bill
30	833500.00	27783.333333



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Thank You